UAP - AI Platform — Architecture

Capabilities, Architecture and Dev Experience

Capabilities, Architecture and Features

Microsoft Agent Framework • Azure Container Apps • Cosmos DB • Azure OpenAI

Unified Platform – UAP, UAIS and UDP

Unified AI Platform components

Unified AI Platform (UAP)

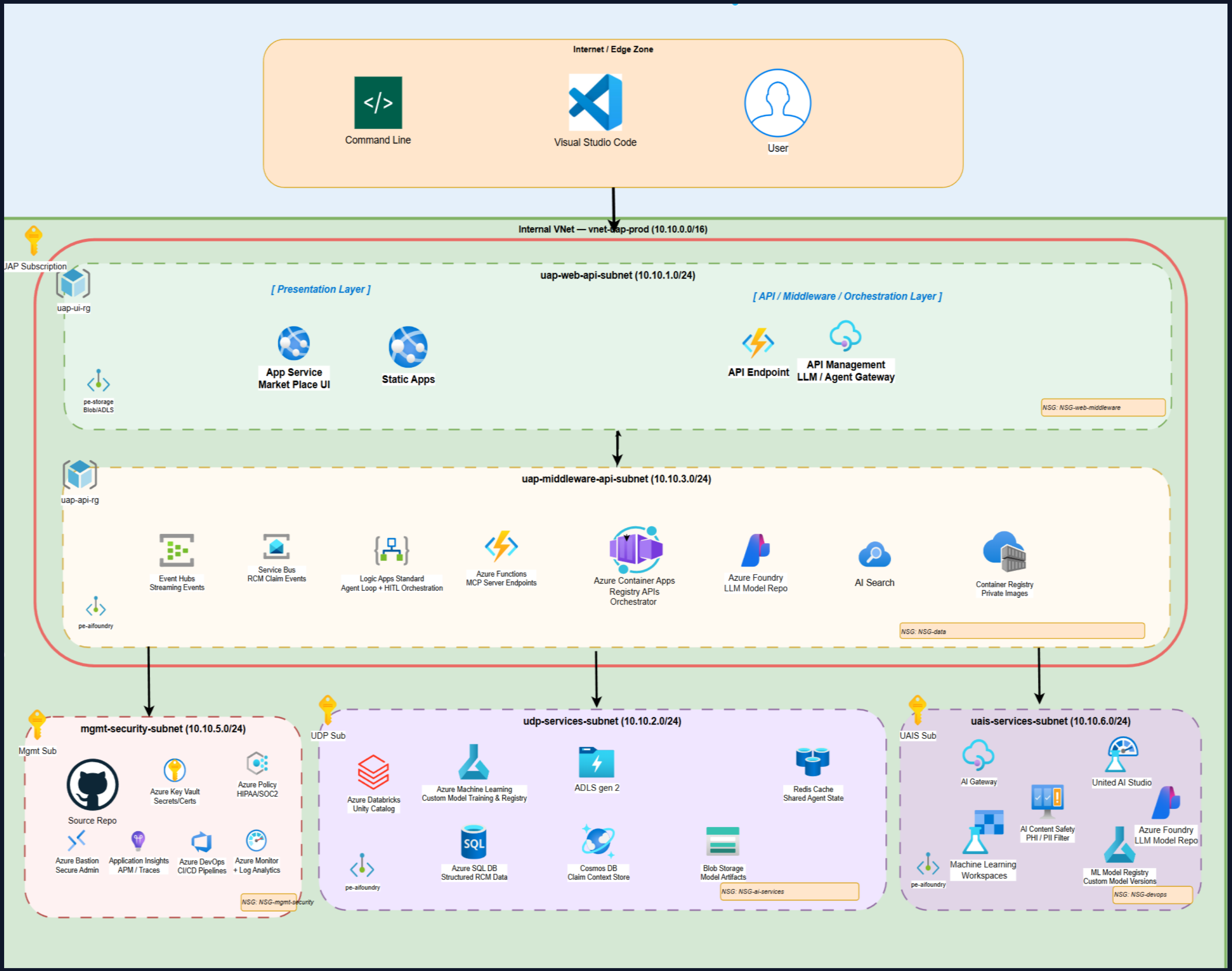
The overarching platform that ties together all AI capabilities — asset catalog, orchestration, governance, and developer experience — into a single, cohesive surface for enterprise teams.

Unified Data Platform (UDP)

The persistence and observability backbone. Cosmos DB (partitioned by tenantId) stores catalog metadata, workflow state, execution history, and audit logs. Application Insights + Azure Monitor supply real-time metrics, distributed traces, and alerting.

Unified AI Services (UAIS)

A middleware layer that normalises access to Azure AI Foundry, Azure OpenAI, MCP servers, and A2A agents. It provides consistent authentication (Entra ID), retry/circuit-breaker policies, and telemetry collection so every upstream consumer sees the same contract.



AI Platform Solution

Unified AI Platform components

Presentation Layer

Next.js 15 frontend served from Azure Container Apps. Includes the Marketplace catalog, Visual Orchestration canvas (React Flow), Publisher Portal, Admin Dashboard, and Playground. Azure Front Door provides edge caching and global routing.

API & Registry Layer

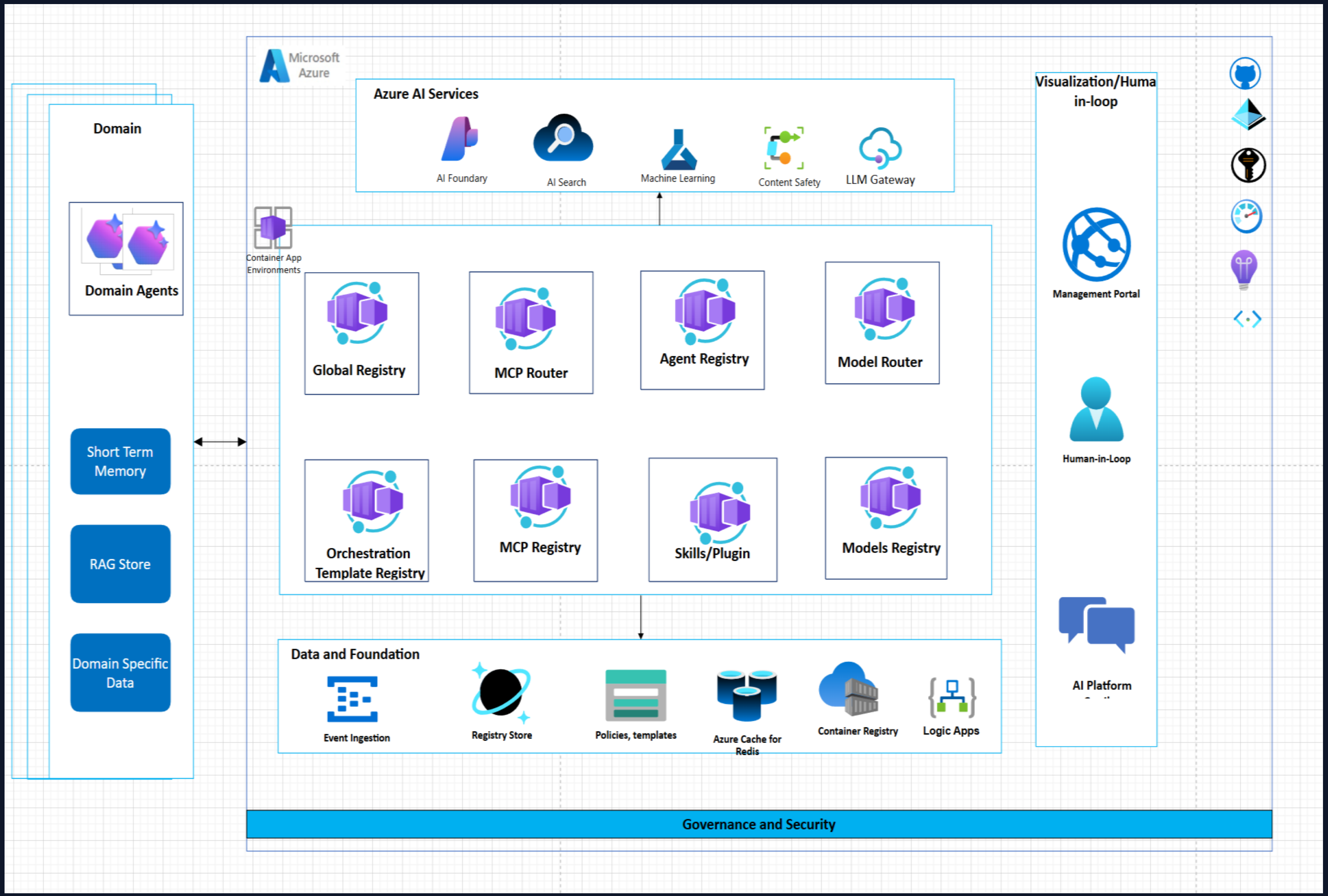
Azure Container Apps(Python/FastAPI) expose RESTful APIs: Asset Catalog CRUD, MCP Server Registry, A2A Agent Registry, Skills Registry, Publisher Submission API, and Security Scanner. Each registry stores data in Cosmos DB with tenantId partition key.

Orchestration & Governance

The Execution Engine (MAF) interprets workflow graphs — dispatching calls to agents, tools, and models. The Policy Engine enforces guardrails (entitlements, risk ratings, budget). Human-in-the-Loop approval gates pause automation for manual review when risk is high.

Azure AI Services

Azure AI Foundry hosts model deployments and agent runtimes. Azure OpenAI provides GPT endpoints. Azure AI Search powers semantic discovery across the catalog. All services integrate via Managed Identity for zero-secret authentication.



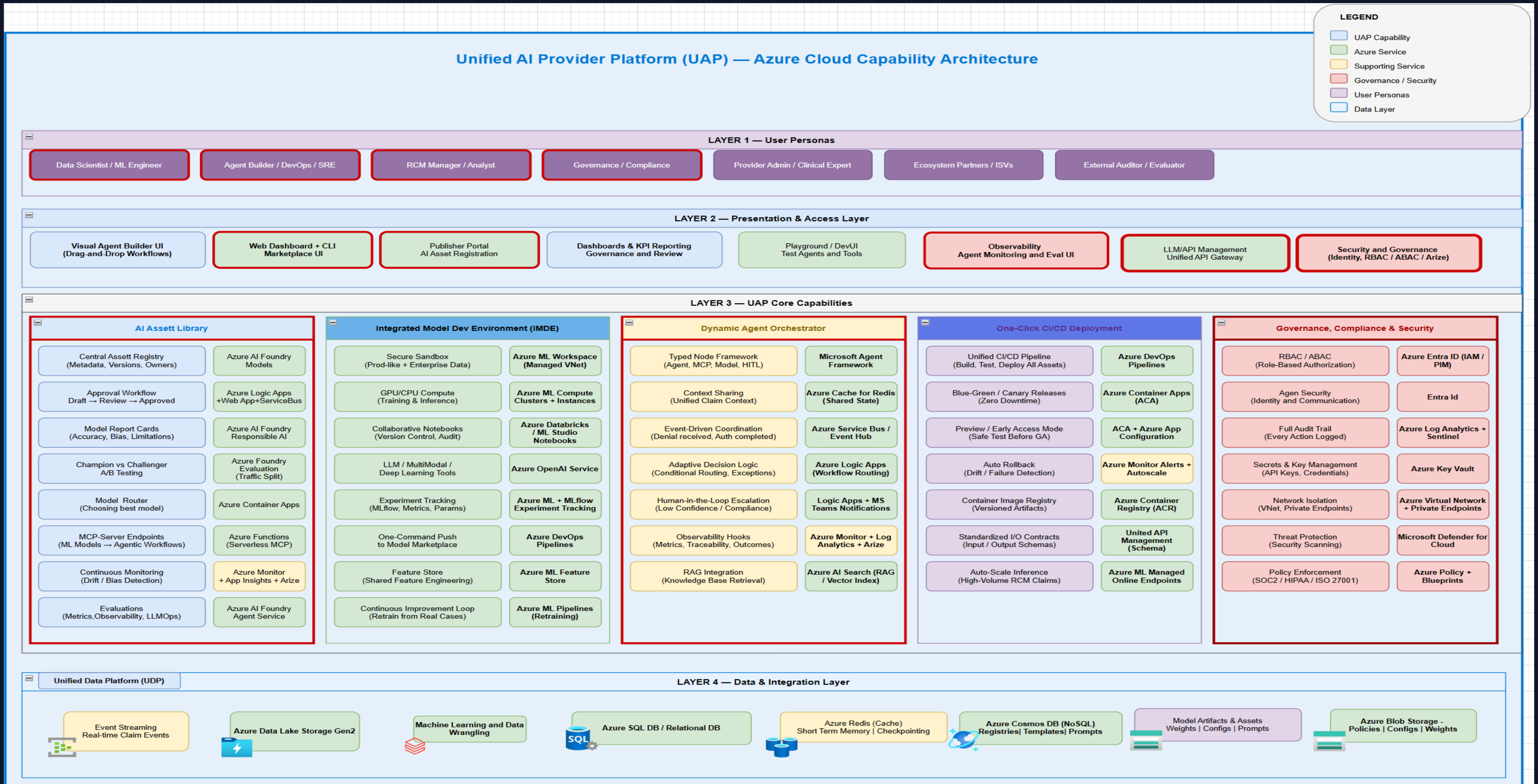
Data & Observability

Cosmos DB is the single NoSQL store (2 MB item limit, hierarchical partition keys). Application Insights + Log Analytics capture distributed traces, custom metrics, and KQL-queryable logs. Redis Cache accelerates hot reads (sessions, lookup data).

Note: Cosmos DB may be replaced with Mongo DB

AI Platform Capabilities

Unified AI Platform components



Note: Capabilites bounded by red boxes are planned to be implemented in next 180 days

Summary of Capabilities — Next 180 Days

22 capabilities across 4 layers targeted for implementation (red-boxed on Slide 4)

LAYER 1 — Presentation & UX

- Visual Orchestration Canvas — Drag-and-drop workflow composition with React Flow
- Admin Dashboard — Governance console — health, submissions, approvals, policy violations
- Publisher Portal — End-to-end asset submission lifecycle: draft → submit → review → publish
- Playground — Interactive sandbox to test agents and tools before deployment
- DevUI — Microsoft Agent Framework debugging and prompt tuning surface

LAYER 2 — API & Registries

- Asset Catalog — CRUD + versioning for agents, MCP servers, tools, models, skills
- MCP Server & Tool Registry — Register/discover MCP endpoints with auth, schemas, health
- Agent Skills Registry — Versioned, governed skill definitions discoverable across workflows
- Publisher & Submission API — Validates metadata/packages, drives review queue and approval states
- Security Scanner — Static + runtime scanning — risky patterns, missing metadata, dependency checks

LAYER 3 — Orchestration & Governance

- Workflow Templates — Reusable orchestration blueprints with node structures and approval steps
- Execution Engine — Graph interpreter — dispatches to agents/tools/models, retries, fan-out/fan-in
- Human-in-the-Loop Gate — Pauses automation for manual review; captures decision provenance
- IAM & RBAC — Entra ID + Managed Identity — role-based view/submit/approve/execute/admin access
- Compensation Logic (Saga) — Multi-step rollback coordination for distributed workflows

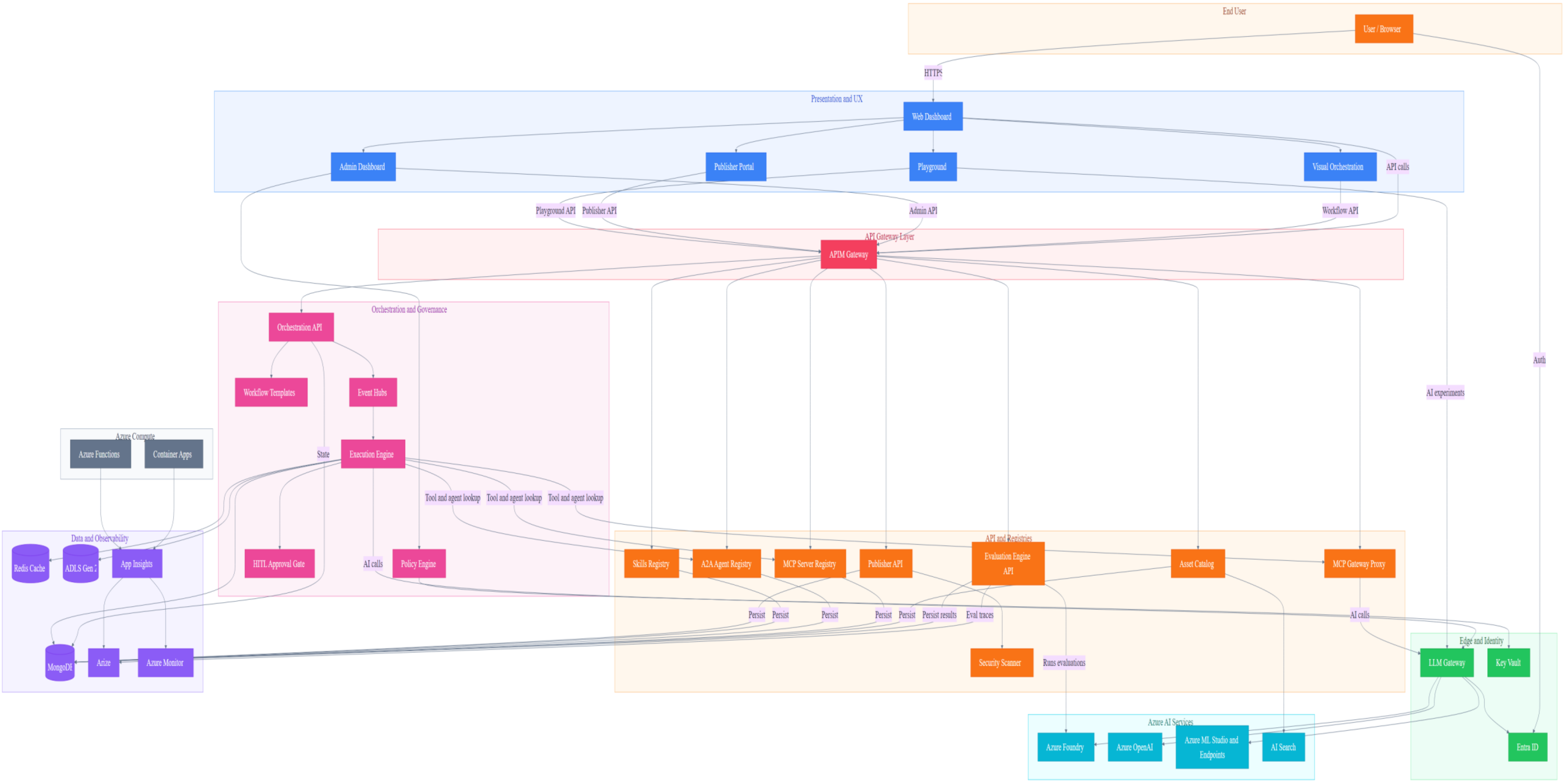
LAYER 4 — Data & Observability

- Audit Log — Immutable records in Cosmos DB with TTL — who did what, when, outcome
- Metrics & OTLP — OpenTelemetry → App Insights — counters, latency, failure rates, cost
- Projects & Workspace — Team-owned work areas with saved workflows, drafts, collaboration
- Ratings & Reviews — Community quality signals for discovery ranking and publisher feedback
- Execution History — Run outcomes, inputs/outputs, trace IDs for debugging and audit

Not in scope: Web Dashboard (already shipping) · Layer 5 Azure Services (provisioned via Bicep IaC independently)

AI Marketplace — App Architecture

User → Web Dashboard (direct) | API Gateway (APIM) in front of registries | LLM Gateway → AI services only | MongoDB replaces Cosmos DB



AI Marketplace — Architecture Features (1–8)

#	Feature	Components	Data Flow	Key Differentiator
1	Enterprise Security	LLM GW, Entra ID, Key Vault, Scanner	LLM GW → Entra (token) → KV (secrets via MI)	Zero-trust, VNet isolation, PHI redaction, HIPAA-ready
2	Plan-Based Orchestration	Execution Engine, Workflow Templates, Registries	Engine → Templates → Registries (bind) → GW Proxy	Declarative plans, Delegation Envelope, step polymorphism
3	Semantic Discovery	AI Search, Asset Catalog, MCP/A2A Registries	Catalog → AI Search (vector) ← NL query → ranked	Vector-indexed search, .well-known crawl, federation
4	Human-in-the-Loop	HITL Gate, Execution Engine, Admin Dashboard	Engine → HITL Gate → Admin UI → Engine (resume)	First-class step type, not bolted on; audit provenance
5	Three-Tier Memory	Redis, Cosmos DB, ADLS Gen 2	Engine → Redis (short) → Cosmos (episodic) → ADLS	Summary-first flush, no raw PHI, episode records
6	Multi-Protocol Tools	MCP GW Proxy, MCP Registry, A2A Registry, Foundry	GW Proxy → MCP servers; A2A Registry → peer agents	MCP + A2A + Foundry + SaaS in one gateway
7	Governance	Policy Engine, Publisher API, Scanner, Admin UI	Publisher → Scanner → Policy → Admin → Catalog	Dual enforcement (publish + execute), full audit trail
8	Observability	App Insights, Azure Monitor, ACA, Functions	ACA + Functions → App Insights → Monitor (alerts)	OpenTelemetry e2e, trace_id propagation, per-episode

AI Marketplace — Architecture Features (9–16)

#	Feature	Components	Data Flow	Key Differentiator
9	Azure-Native Not Locked	MCP GW Proxy, A2A Registry, Foundry, ACA	GW routes to any MCP (ACA, SaaS, on-prem)	Open protocols, federation, optional Foundry
10	Production Infra	ACA, Functions, EventHub, LLM Gateway, Cosmos DB	ACA (stateless) + Functions (event) + LLM Gateway (edge)	Subnet isolation, scale-to-zero, Bicep IaC
11	Intelligent Tool Finder	AI Search, MCP Registry, Azure OpenAI	Registry → Cosmos → AI Search (FAISS) ← NL query	FAISS vector matching — agents self-discover tools
12	Dynamic Invocation	Execution Engine, MCP GW Proxy, AI Search	Engine → Search (find) → Registry (schema) → GW	Discover → resolve → call in one turn, no pre-config
13	Load Balancing	LLM Gateway, ACA, MCP GW Proxy	LLM GW → GW Proxy → ACA (scaling + probes)	Health-aware routing across MCP server replicas
14	IDE Integration	MCP Registry, Asset Catalog, LLM Gateway	Registry → export mcp.json → VS Code / Cursor	One-click IDE setup from marketplace registry
15	JWT Token Vending	Entra ID, Key Vault, LLM Gateway	Agent → Entra (creds) → JWT → LLM GW → GW Proxy	M2M scoped tokens for autonomous agent auth
16	Context-Aware Perms	Policy Engine, Entra ID, LLM Gateway	Engine → Policy (claims + context) → allow/deny	Dynamic scope eval per invocation, not just RBAC

General Orchestrator Workflow

#	Step	Component	Description
1	Client Request	LLM Gateway	Request enters via domain LLM Gateway with Entra auth
2	Classify Intent	Azure Foundry /classify	Determine intent, phase, and domain routing
3	Governance Pre-flight	Policy Engine + Entra	Check entitlements, PHI class, budgets, HITL rules
4	Select Domain + Plan	AI Search + Registries	Query plan registry and domain orchestrator registry
5	Resolve + Bind Tools	MCP/A2A Registries	Bind agents via A2A, tools via MCP from registries
6	Delegation Envelope	Orchestration API	Create least-privilege envelope with plan + constraints
7	Domain Execution	Event Hubs + Engine	Fan-out via Event Hubs, step loop in Execution Engine
8	Memory + Evidence	Redis + Cosmos DB	Short-term state in Redis, episodic memory in Cosmos
9	Post-flight Checks	Content Safety + Policy	PHI redaction, compliance, channel HITL
10	Normalize + Respond	LLM Gateway	Standard schema + provenance + trace_id, return result

Global orchestrator manages intent classification, governance, plan selection, domain delegation, and post-flight normalization

LangGraph vs MAF — Capability Comparison

#	Capability	LangGraph	MAF	Rebuild in LangGraph
1	Graph-based orchestration	StateGraph DAG	WorkflowBuilder DAG	—
2	Central shared state	StateGraph	Intentionally absent	Session/state isolation
3	Step-level checkpointing	Checkpointers	Workflow checkpoints	—
4	Pause / resume execution	Via checkpoints	Via workflow runtime	—
5	Time-travel / replay	Checkpoint history	Partial (telemetry)	Full replay tooling
6	Agent statelessness	Stateful by design	Enforced	State discipline
7	Typed message contracts	Dict-based	Strong typing	Validation + schema
8	Workflow execution waves	Super-steps	Supersteps	—
9	Human-in-the-loop	Interrupts + edits	First-class	—
10	Built-in observability	Requires LangSmith	OTel native	Azure telemetry
11	Workflow-level spans	No	Executor spans	Deep tracing
12	Azure Monitor integration	No	Native	Full APM
13	Managed identity support	No	Native	Identity mgmt
14	Compliance logging	No	Native hooks	Audit trail
15	Dev / prod parity	Platform-dependent	Designed-in	Env isolation
16	Enterprise RBAC	No	Platform-level	RBAC enforcement

LangGraph excels at graph orchestration; MAF leads on enterprise compliance, observability, and Azure integration

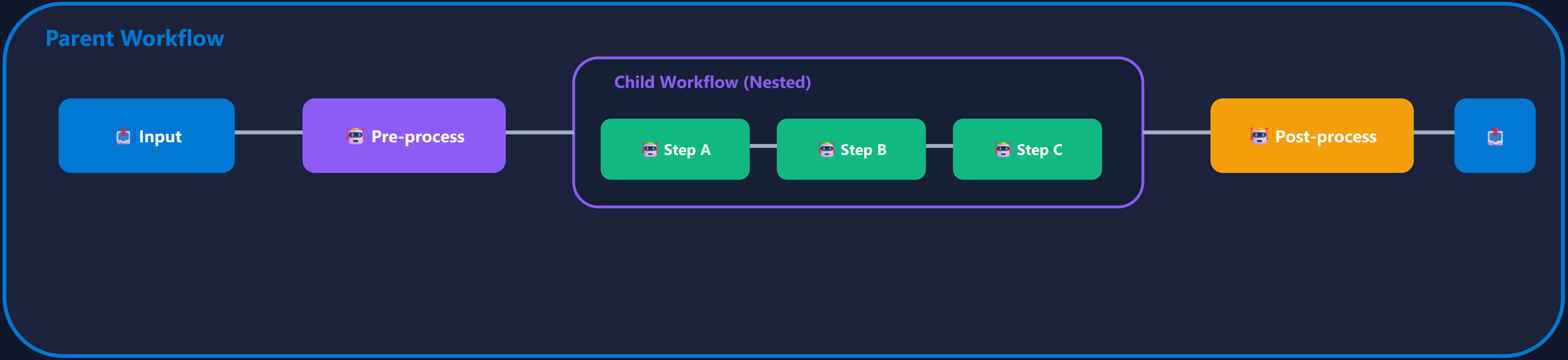
LangGraph vs MAF — Compliance Comparison

#	Requirement	LangGraph on Azure	MAF on Azure
1	PHI / PII handling	Custom-built	Native hooks
2	Audit trails	LangSmith / custom	Native workflow events
3	Deterministic replay	Yes	Partial
4	Execution provenance	Graph-level	Executor-level
5	Role-based access	App-level	Platform-level
6	Azure AD integration	Manual	Native
7	Managed Identity	Manual	Native
8	Secrets management	App-managed	Platform-managed
9	Logging retention	Custom	Azure Monitor
10	Incident forensics	Hard	Straightforward
11	Security review friction	High	Low

Side-by-side comparison for enterprise healthcare compliance requirements

1 Global Orchestrator Pattern

Compose workflows hierarchically — a domain workflow as a step in a parent workflow



When to Use

- Reusable workflow modules (e.g., ‘claim check' sub-workflow)
- Team-owned sub-workflows composed into larger pipelines
- Isolate complex logic into testable sub-units
- Recursive workflows: a workflow can contain itself
- A2A: cross-service workflow composition

MAF Code Pattern

```
# Sub-workflow as executor
child_wf = Workflow()
child_wf.add_executor('step_a', agent_a)
child_wf.add_executor('step_b', agent_b)
child_wf.add_edge('step_a', 'step_b')

parent_wf = Workflow()
parent_wf.add_executor('preprocess', pre_agent)
parent_wf.add_executor('child', child_wf.as_executor())
parent_wf.add_edge('preprocess', 'child')
```

Key Characteristics

- MAF: sub_workflow_basics — wrap Workflow as executor via .as_executor(); supports kwargs propagation, request interception, checkpointing

AI Platform — Sequence Diagrams

Developer Experience & RCM Domain Orchestration Flows

12 Sequence Diagrams • 6 Dev-Experience • 6 RCM Domain Workflows

Microsoft Agent Framework • Azure Container Apps • Cosmos DB • Azure OpenAI

Appendix – Work in Progress

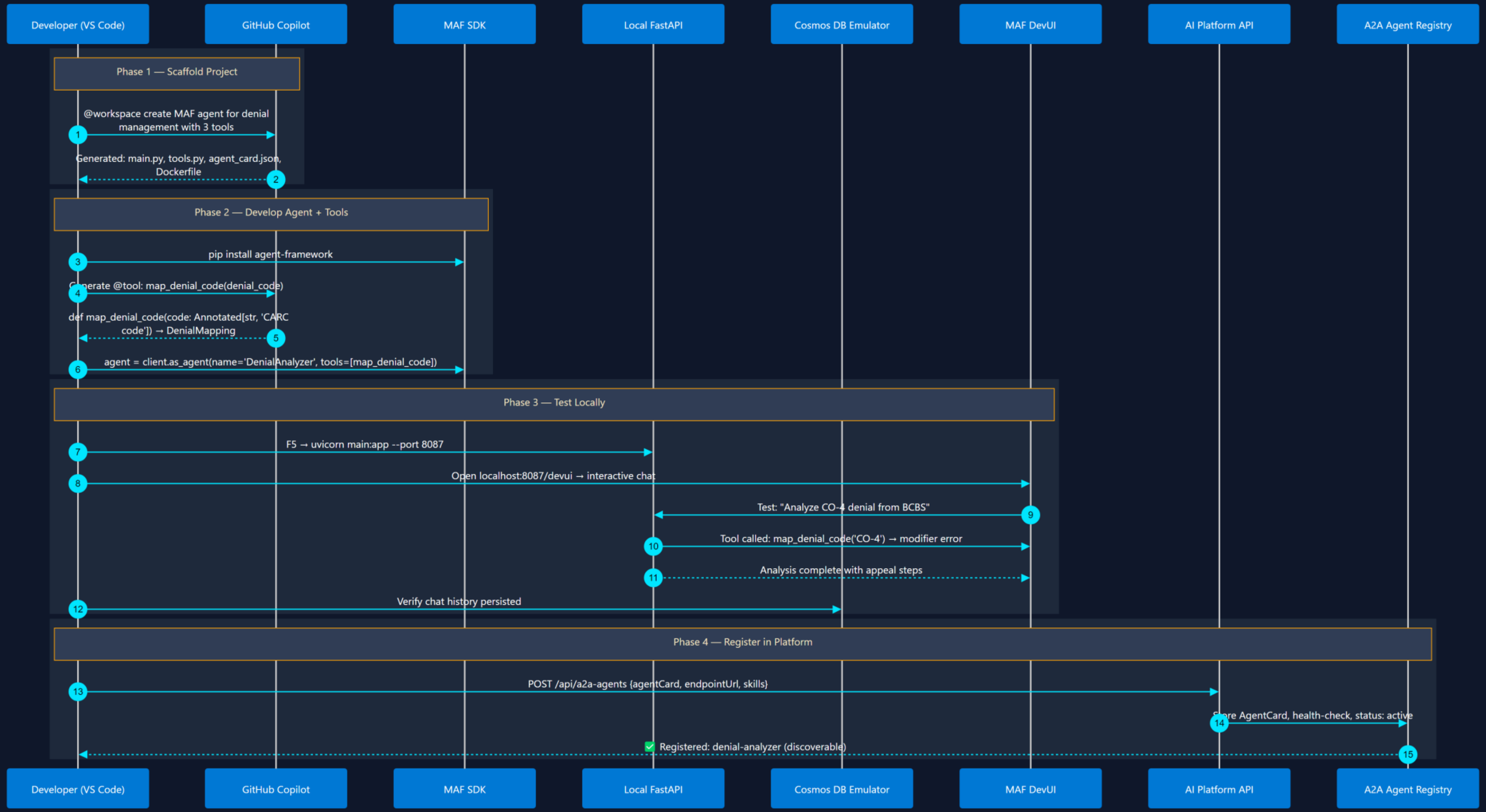
Microsoft Agent Framework • Azure Container Apps • Cosmos DB • Azure OpenAI

1

Agent Development Lifecycle

From empty folder to a registered, discoverable A2A agent

Scaffold → Code → Test → Register



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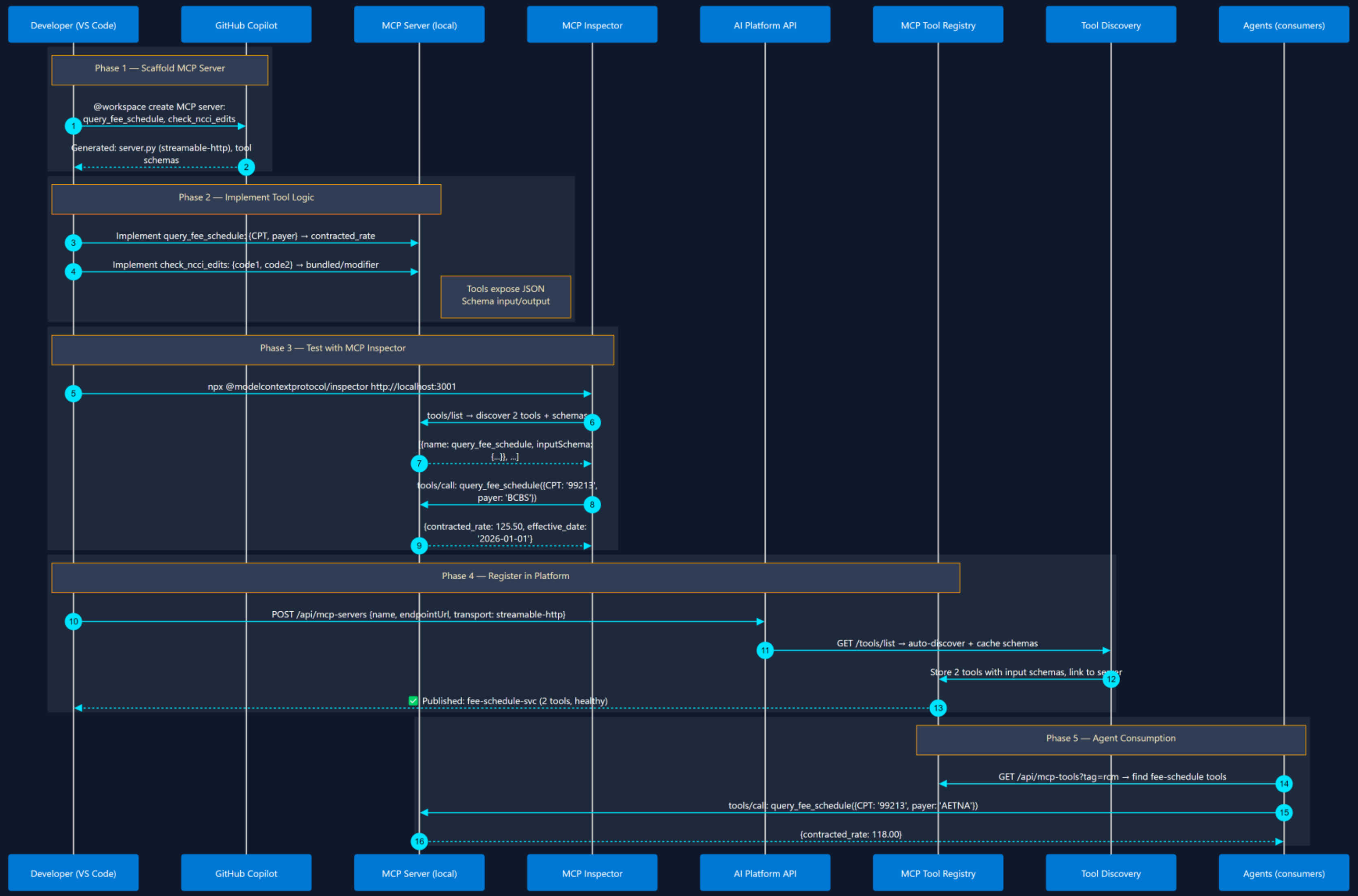
MCP Tool Development & Registration

Build → Test → Publish → Discover

Build tool server → test schema → publish to registry → agents discover it



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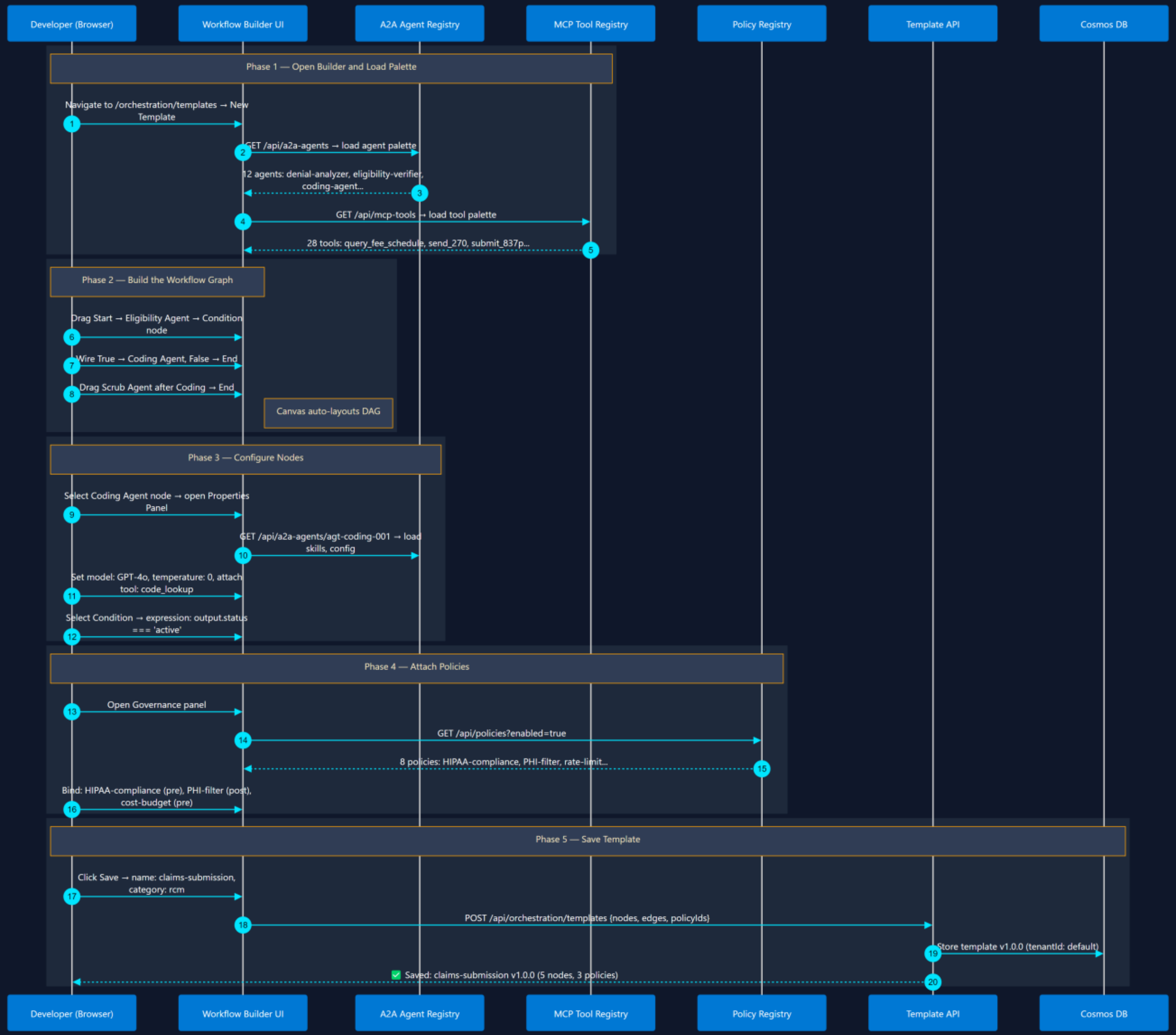
Orchestration Template Design

Visual workflow builder → select agents → wire edges → attach policies → save



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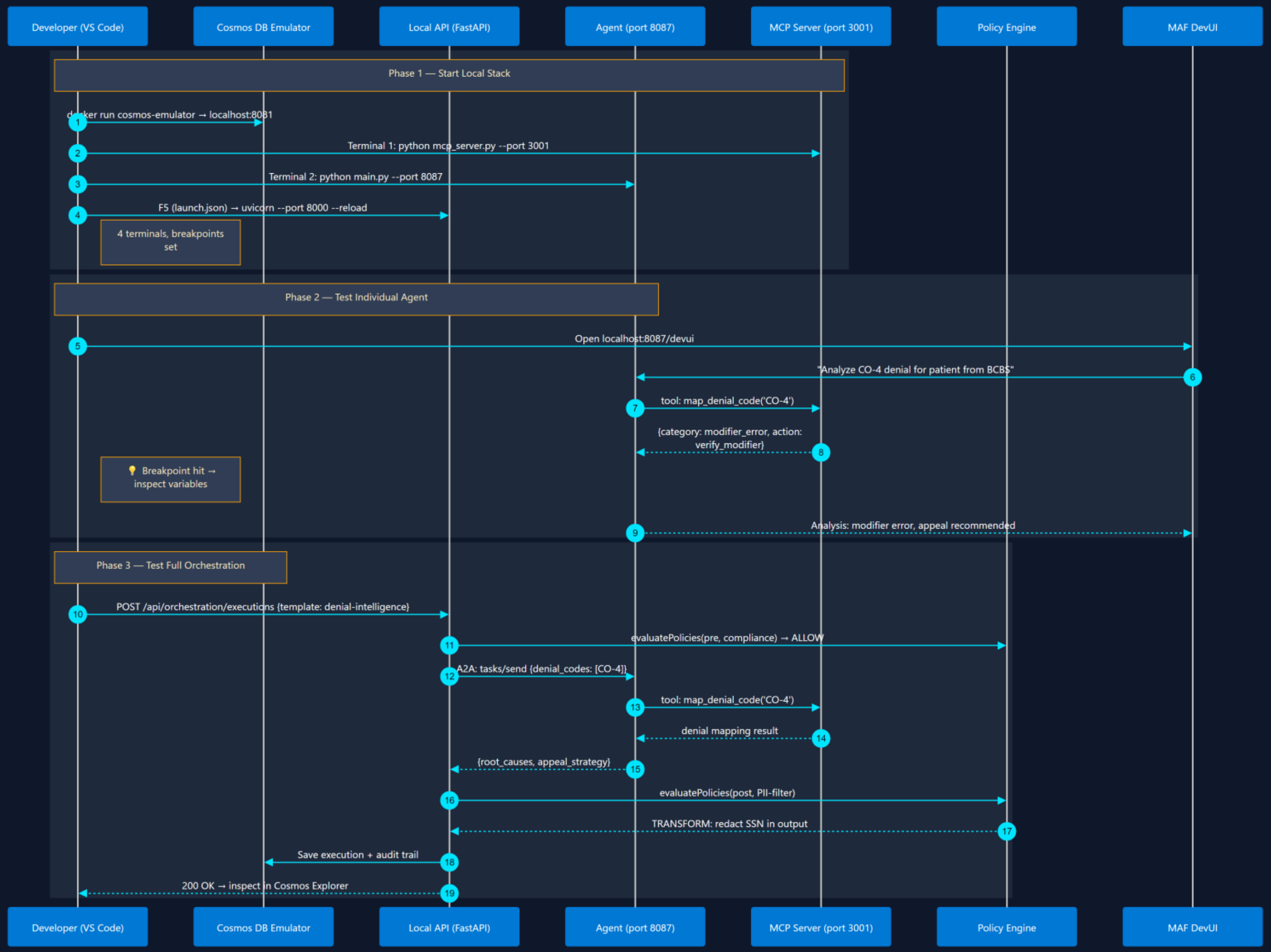
Local Testing & Debugging

Cosmos Emulator → F5 debug → test orchestration → inspect policy decisions

VS Code Debug + Emulators



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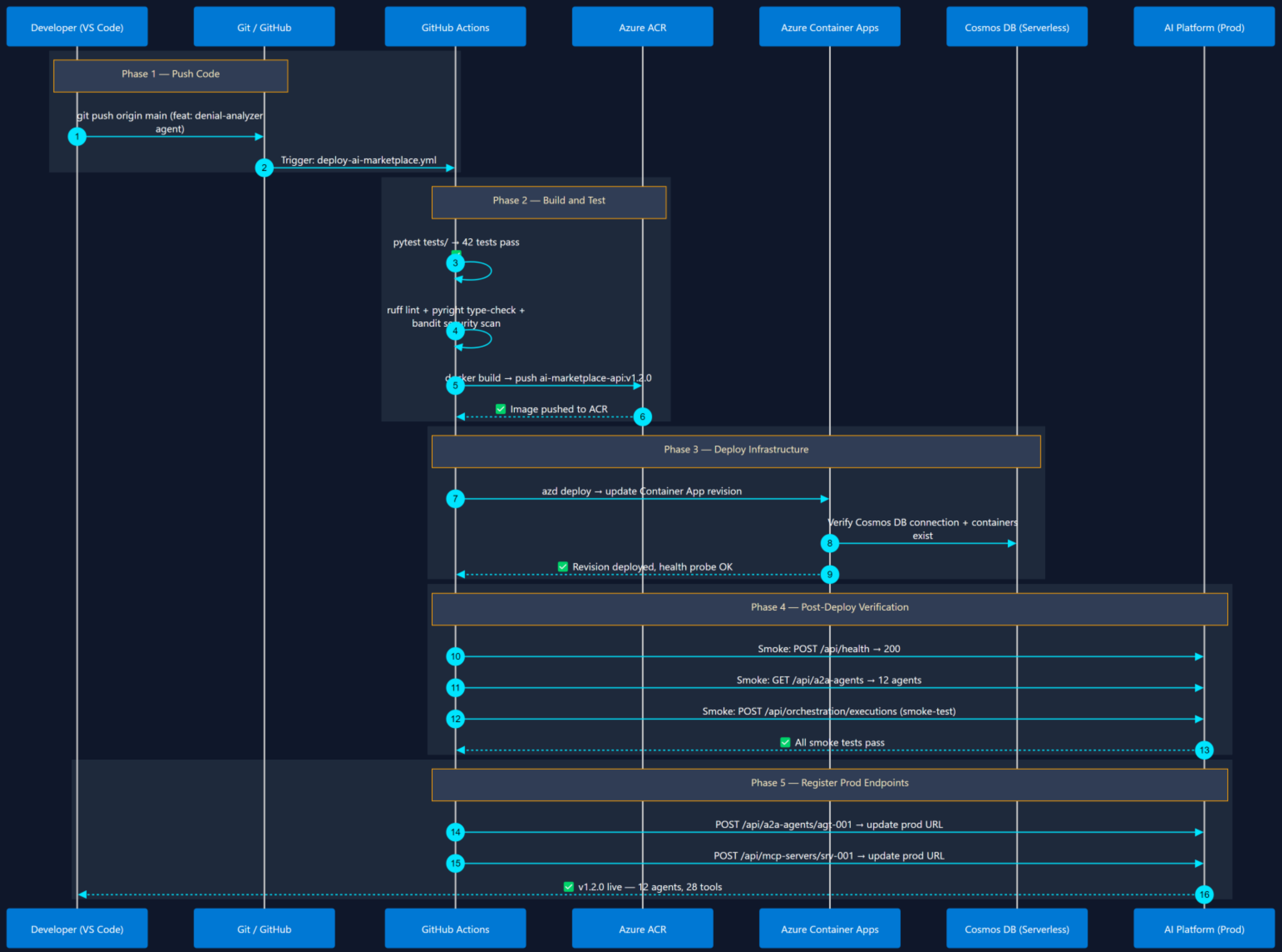
CI/CD Pipeline & Deployment

azd up + GitHub Actions

Git push → GitHub Actions → build container → deploy to Container Apps → verify



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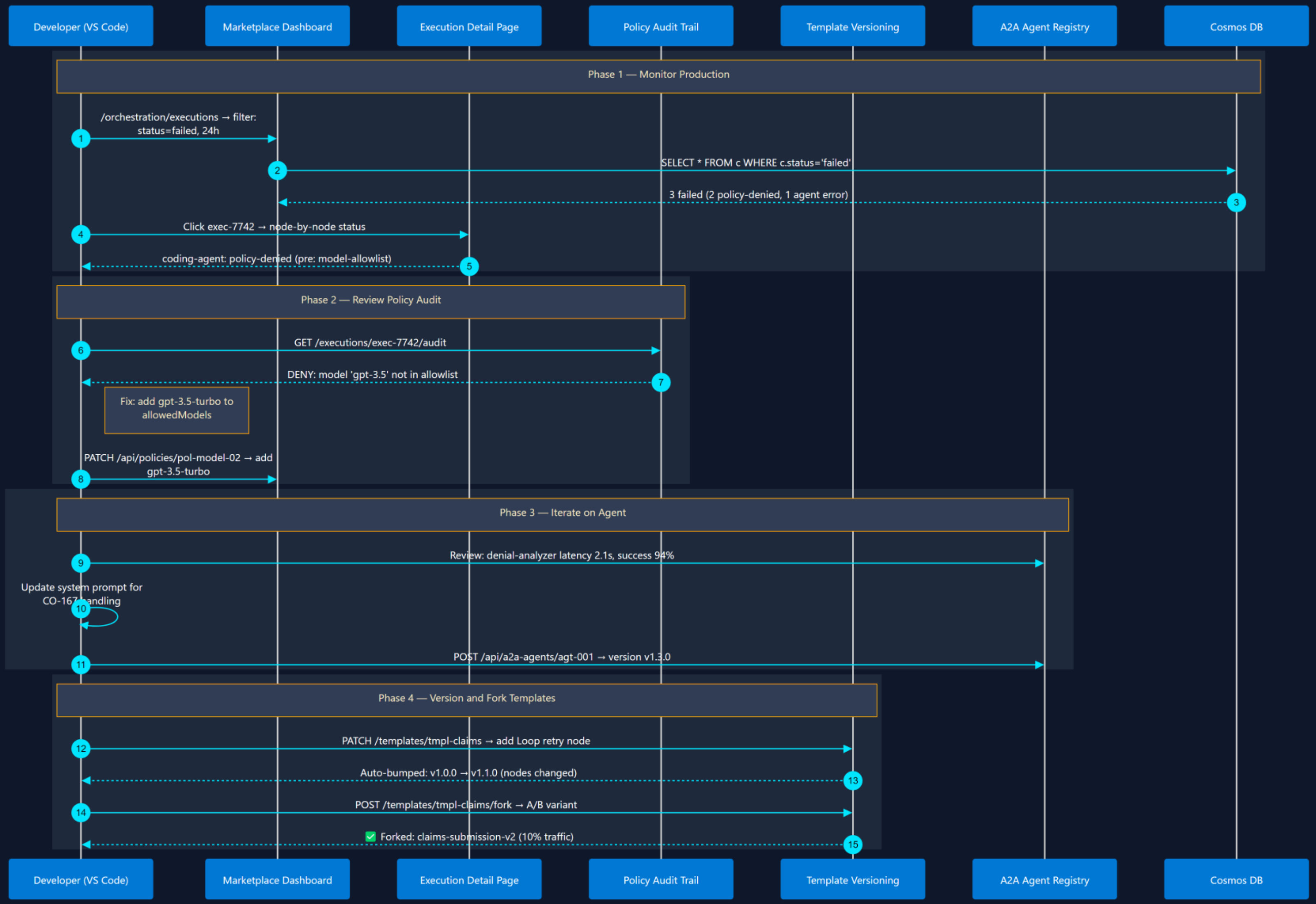
Day-2: Monitor, Iterate & Version

Observe → Refine → Ship

Monitor executions → review policy violations → update agents → version templates



180-Day Implementation Roadmap — Red Bounding Boxes

Slide 4 diagram: Red-outlined components are targeted for the next 180-day implementation cycle

■ = Targeted for 180-day implementation

□ = Already built or out of scope for this cycle

GROUP 1 — Core Platform (Left Red Boxes)

Layer 1 — Presentation & UX

- Web Portal /CLI (Discovery(React Flow drag-drop)
- Admin Dashboard (governance, health, approvals)

Layer 2 — API & Registries

- AI Asset Catalog (CRUD, versioning, search)
- MCP Server & Tool Registry

Layer 3 — Orchestration & Governance

- Workflow Templates (reusable patterns)
- Execution Engine (graph dispatch, retries, fan-out)

Layer 4 — Data & Observability

- Audit Log (immutable, TTL, Cosmos DB)
- Metrics & OTLP (OpenTelemetry → App Insights)

GROUP 2 — Governance & Publisher (Right Red Boxes)

Layer 1 — Presentation & UX

- Publisher Portal (submission lifecycle)
- Playground (interactive agent/tool testing)
- DevUI (MAF agent debugging surface)

Layer 2 — API & Registries

- Agent Skills Registry
- Publisher & Submission API
- Security Scanner (static + runtime validation)

Layer 3 — Orchestration & Governance

- Human-in-the-Loop Approval Gate
- IAM & RBAC Access Control (Entra ID)
- Compensation Logic / Saga (rollback coordination)

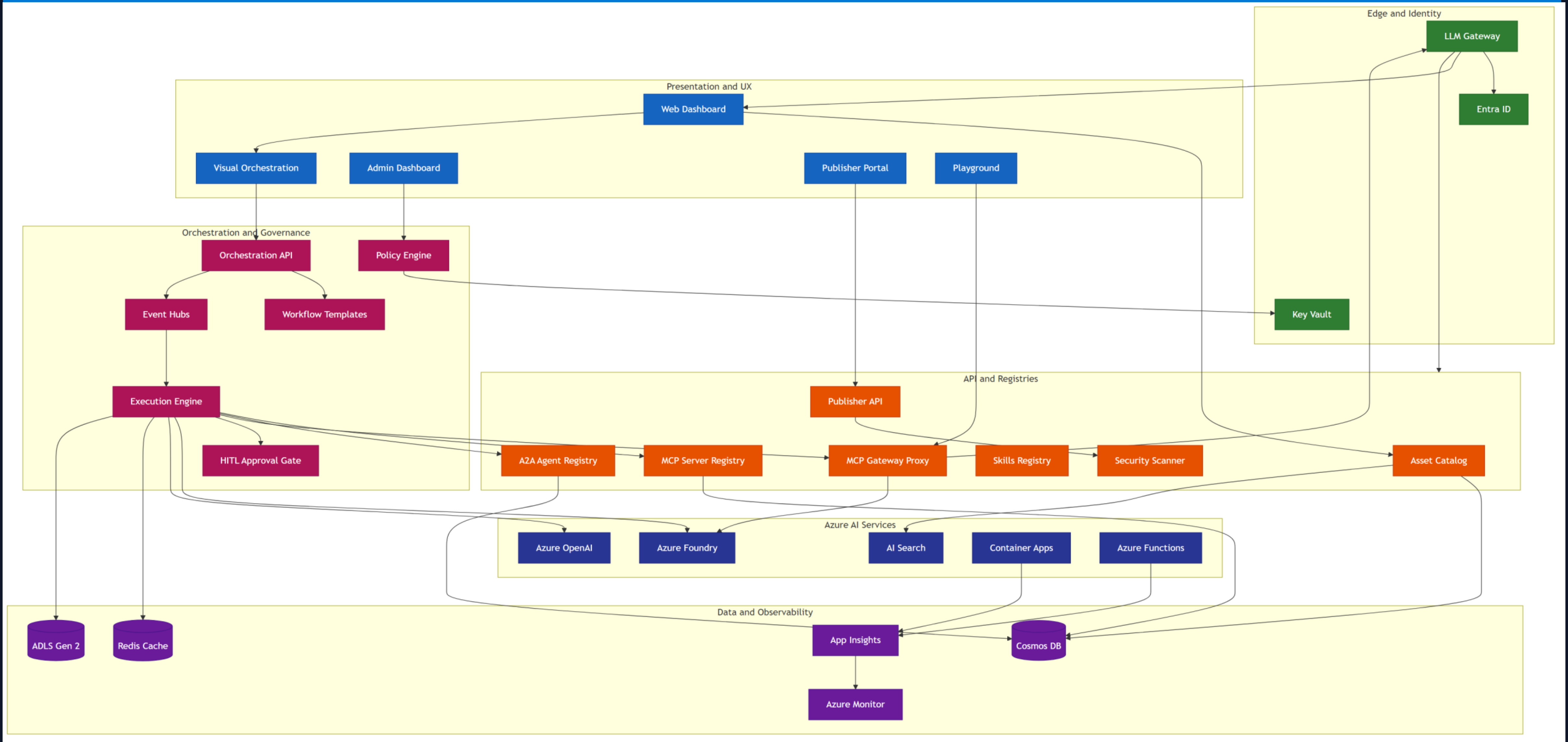
Layer 4 — Data & Observability

- Projects & Workspace (team-owned work areas)
- Ratings & Reviews (community feedback)
- Execution History (run outcomes, traces)

Not red-boxed: Web Dashboard (already shipping) • Layer 5 Azure Services (provisioned via Bicep IaC independently)

AI Marketplace app architecture

Enterprise grade scalable platform



Next Steps— Quick Reference

All 10 patterns are available today in Microsoft Agent Framework (Python + .NET) • Hosted on Azure Container Apps • Cosmos DB state • Azure OpenAI